

CS4100 Report

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We set out to compare two explanation models—L2X and SHAP—on a logistic-regression model we trained ourselves. Our goal was to see which pixels each method deems most “important” when classifying MNIST handwritten digits. MNIST’s simplicity makes it ideal for visually evaluating each explainer’s output.

Concrete Technical Problem Statement

Demonstrate and compare instance-wise interpretability on a logistic regression classifier for the MNIST dataset:

- **Inputs:** Gray-scale digit images (28×28 pixels)
- **Outputs:** (1) Predicted digit (0–9) and (2) pixel-level importance scores

Dataset Description

MNIST consists of 70,000 gray-scale, 28×28 images of handwritten digits—60,000 for training and 10,000 for testing. It’s a standard benchmark for image classification due to its clarity and uniform size.

Methods and Algorithms

We applied two post-hoc explainers:

1. **L2X (Learning to Explain)** A neural selector that maximizes mutual information between chosen pixels and model predictions, selecting the top-K most informative pixels. Such that mutual information is defined as:

$$I(X;Y) = \mathbb{E}_{X,Y} \left[\log \frac{p_{XY}(X,Y)}{p_X(X)p_Y(Y)} \right]$$

where:

$I(X;Y)$ Mutual information between the random variables X and Y , i.e. the amount of information shared by X and Y .

$\mathbb{E}_{X,Y}[\cdot]$ Expectation taken with respect to the joint distribution $p_{XY}(x,y)$.

$p_{XY}(X,Y)$ Joint probability density (or mass) of X and Y .

$p_X(X), p_Y(Y)$ Marginal densities of X and Y , respectively.

$\log \frac{p_{XY}(X,Y)}{p_X(X)p_Y(Y)}$ Pointwise information gain: how much more likely the pair (X,Y) is under the joint distribution than if X and Y were independent.

In the L2X method, we train a selector network to choose a subset of features $S \subseteq \{1, \dots, d\}$ by maximizing

$$I(X_S;Y) = \mathbb{E}_{X_S,Y} \left[\log \frac{p_{X_S,Y}(X_S,Y)}{p_{X_S}(X_S)p_Y(Y)} \right],$$

thereby encouraging the selector to pick those input features (pixels) that carry the most information about the model’s prediction Y .

2. **SHAP** A game-theoretic, model-agnostic framework that assigns each pixel a contribution score via permutation or gradient methods, visualized as a continuous heatmap such that.

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} [f(x_{S \cup \{i\}}) - f(x_S)]$$

where

ϕ_i Shapley value (contribution) of feature i to the model output.

N The set of all feature indices, $|N| = n$.

$S \subseteq N \setminus \{i\}$ Any subset of features not containing i .

$|S|!(n - |S| - 1)!/n!$ Weighting factor: the fraction of all possible coalitions of size $|S|$ in which feature i could join.

$f(x_{S \cup \{i\}})$ Model prediction when only features in S plus feature i are present.

$f(x_S)$ Model prediction when only features in S are present.

$f(x_{S \cup \{i\}}) - f(x_S)$ Marginal contribution of feature i to coalition S .

By visualizing $\{\phi_i\}$ as a heatmap over input pixels, SHAP highlights exactly how much each pixel “pushes” the prediction up or down relative to a baseline.

Using a simple logistic-regression baseline made it easier to isolate differences in the explainers’ behaviors.

Relation to Existing Work

This project builds on:

- Chen et al., “Learning to Explain” (L2X, 2018)
- Lundberg & Lee, “A Unified Approach to Interpreting Model Predictions” (SHAP, 2017)

Unlike purely theoretical or black-box evaluations, we perform an empirical, side-by-side comparison on a model we trained ourselves.

Experimental Setup and Results

Model training: Logistic regression on MNIST with learning rate = 0.01, epochs = 10. **Explanation parameters:**

- **L2X:** Top 50 pixels selected
- **SHAP:** Gradient-based explainer on model outputs

Hypotheses: L2X will highlight coherent clusters corresponding to strokes, while SHAP will produce more granular pixel-wise attributions.

Findings

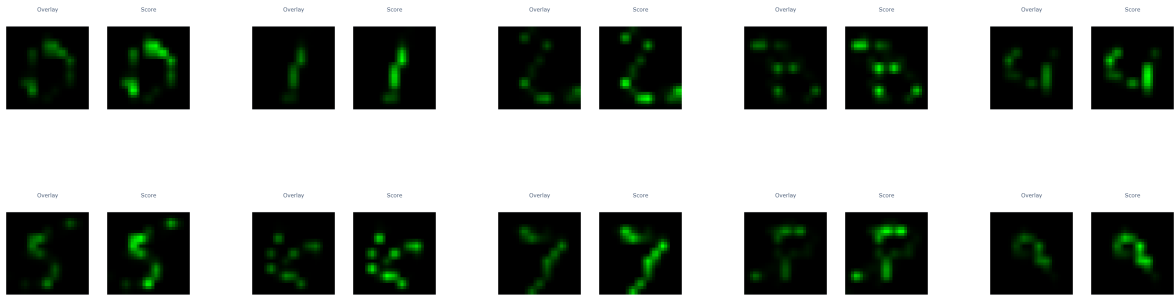


Figure 1: L2X-selected pixel masks for digits 0–9.



Figure 2: SHAP attribution heatmap for a sample MNIST digit.

These are more easily visible on GitHub:

<https://github.com/WilsonGlass/Explainers/blob/master/exec.ipynb>

Comparing Digits

- 0 SHAP focuses on the pixels *present* in the digit, whereas L2X attends strongly to the *absence* of pixels at the center—an informative cue for classifying a “0.”
- 1 The explanations look similar overall, but L2X additionally notices the empty band of pixels surrounding the vertical stroke.
- 2 SHAP assigns little importance to the top curve of the “2,” while L2X highlights it.
- 3 SHAP pinpoints a handful of specific pixels; L2X spreads its attention across the whole shape.
- 4 SHAP values the intersection of the two strokes; L2X emphasizes the blank space between the upper strokes—absence again serving as signal.
- 5 SHAP concentrates on the left curve. L2X highlights the upper segment, though its pattern is less interpretable here.
- 6 SHAP’s attribution appears almost random, whereas L2X crisply outlines the inner and upper curves—perhaps the clearest L2X explanation in the set.
- 7 SHAP covers most of the digit, offering limited insight. L2X singles out the acute corner at the top right.
- 8 SHAP rates both lobes and the waist as important; L2X focuses on the central voids inside the two loops.
- 9 SHAP highlights the curved hook at the top; L2X captures that region too, but its mask is less conclusive.

Overall SHAP tends to emphasize *inked* pixels, while L2X often derives information from *where ink is absent*. The contrast underscores why consulting multiple explanation methods can give a fuller picture of model reasoning.

What Worked and What Didn’t

Although the overall results are encouraging, the visualization pipeline was unexpectedly troublesome. Our initial goal was to render both explainers with a uniform heat-map color scheme, but matching palettes and scales proved more effort than it was worth—and, in retrospect, might have obscured meaningful differences.

In the end we retained each explainer’s default rendering. This choice preserves the fullest signal each method provides, even if it sacrifices cosmetic uniformity. We therefore rely on our own side-by-side inspection to interpret the contrasting attributions, rather than forcing them into an artificially identical visual style.

Discussion and Future Directions

Challenges

- Tuning each explainer’s hyperparameters to achieve stable, comparable outputs.
- Managing and resolving dependency conflicts between libraries.
- Reshaping images into different dimensions for the explainers and then back for display.
- Integrating the explainers with our model’s existing codebase.
- Presenting the resulting explanation images in a consistent, side-by-side format for fair comparison.

The project proved more challenging than expected. While libraries for both **L2X** and **SHAP** exist, their dependency stacks did *not* play nicely together; we spent more time wrestling with version conflicts than writing new code.

Compounding the issue, the logistic-regression model we trained in class was *not* designed with explanation frameworks in mind, so we had to insert several ad-hoc work-arounds to make the explainers run.

Recommendation. If a future team pursues a similar comparison, start with an established, explanation-ready model (e.g. a pre-trained CNN packaged with Captum or OmniXAI) instead of one built from scratch. This shift would let the group focus on the interpretability questions rather than environment troubleshooting.